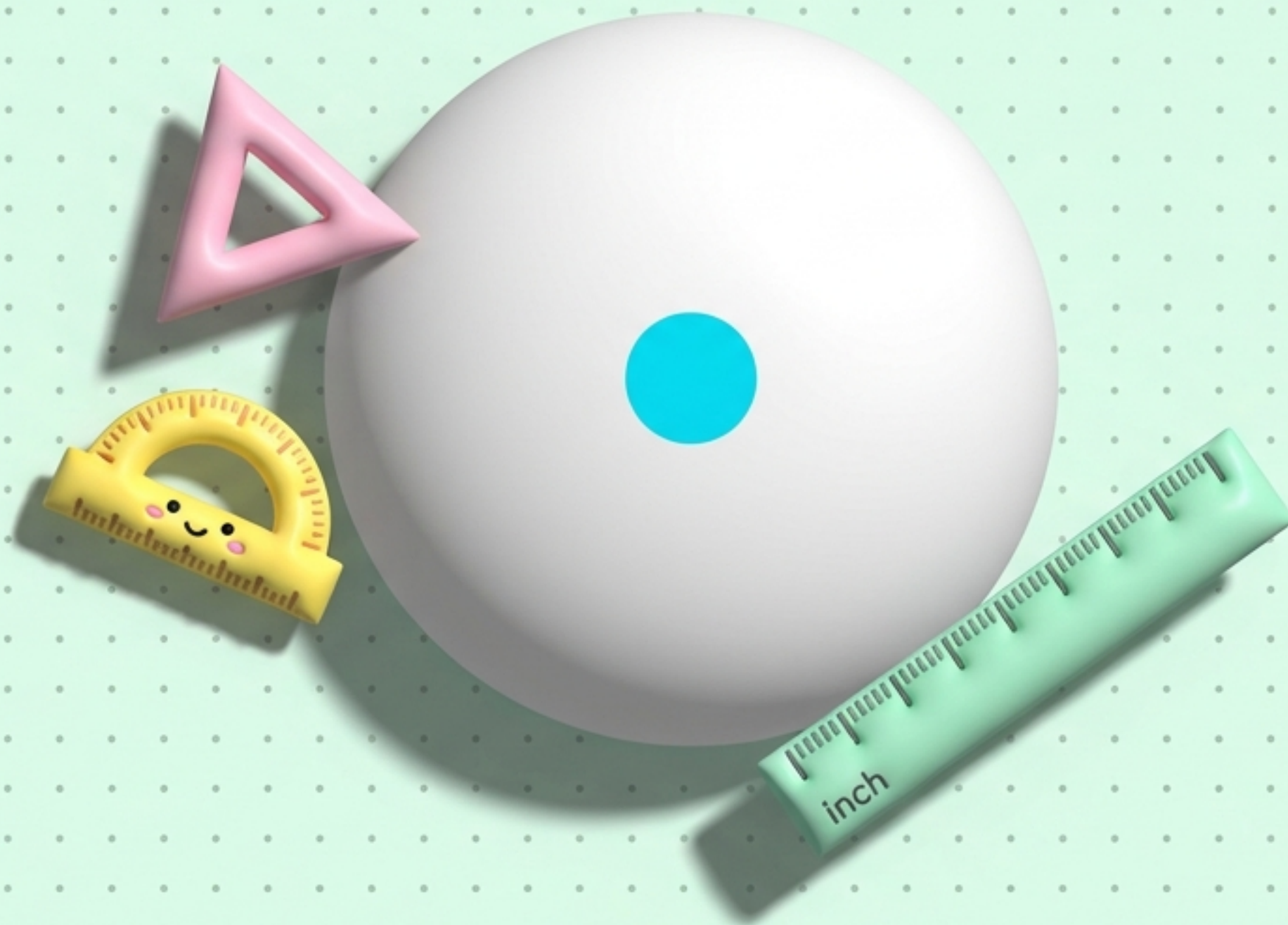


# Class 10 Math: Circles

## *The Ultimate Visual Study Guide*

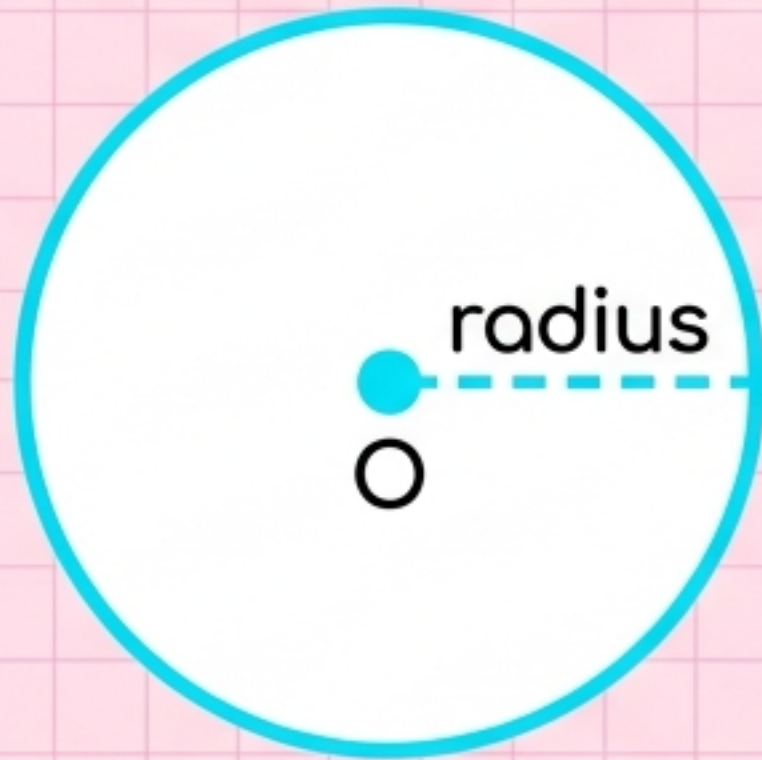
Chapter 10  
Revision





*What happens when these two meet in the same plane?*

## The Circle



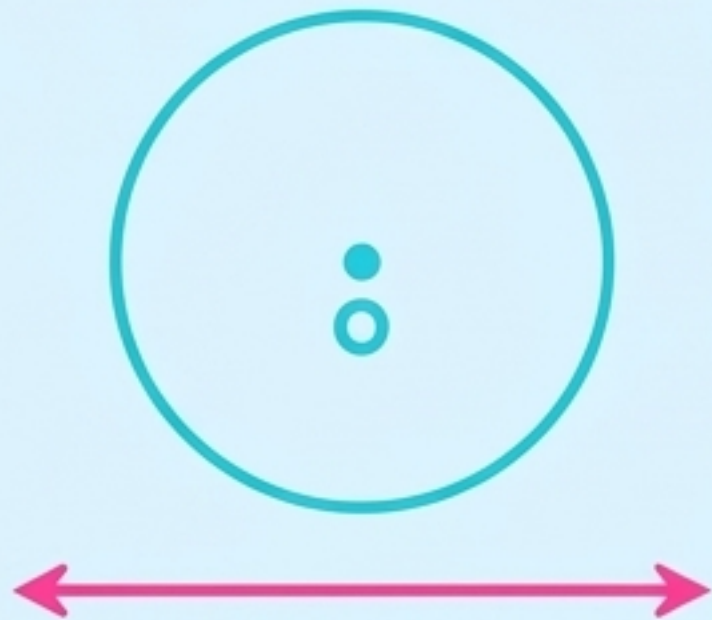
A collection of all points in a plane at a constant distance (radius) from a fixed point (centre).

## The Line

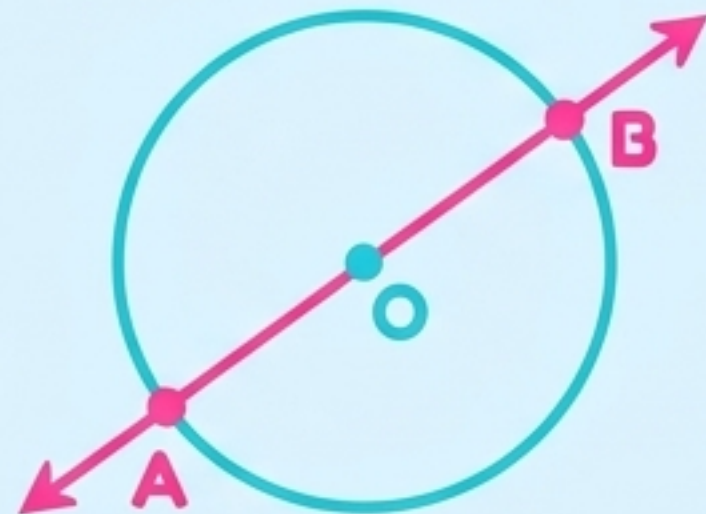


A straight, one-dimensional figure having no thickness and extending infinitely in both directions.

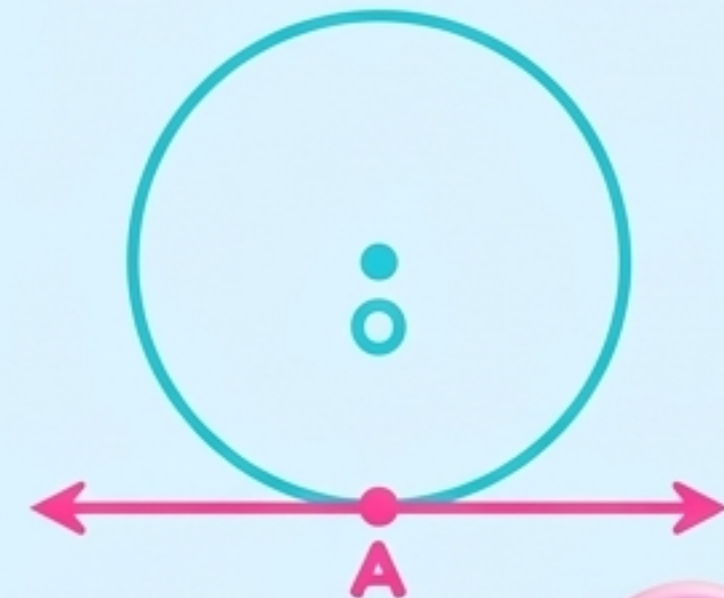
# The 3 Possibilities



Non-intersecting Line  
**0 Common Points**

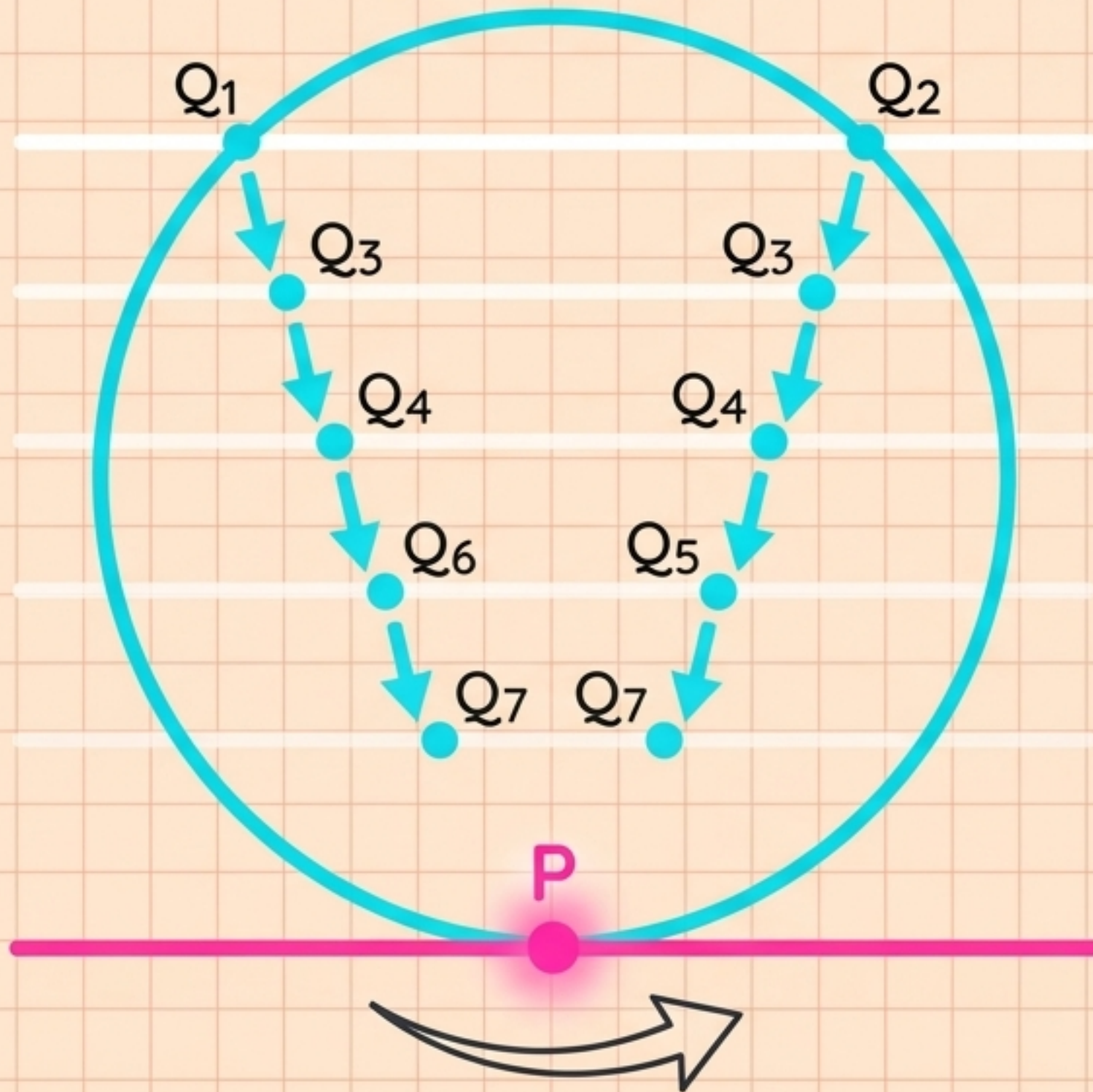


Secant  
**2 Common Points  
(A & B)**



Tangent  
**1 Common Point (A)**

# Anatomy of a Tangent



## Golden Rule:

The tangent is a special case of the secant, when the two end points of its corresponding chord coincide!

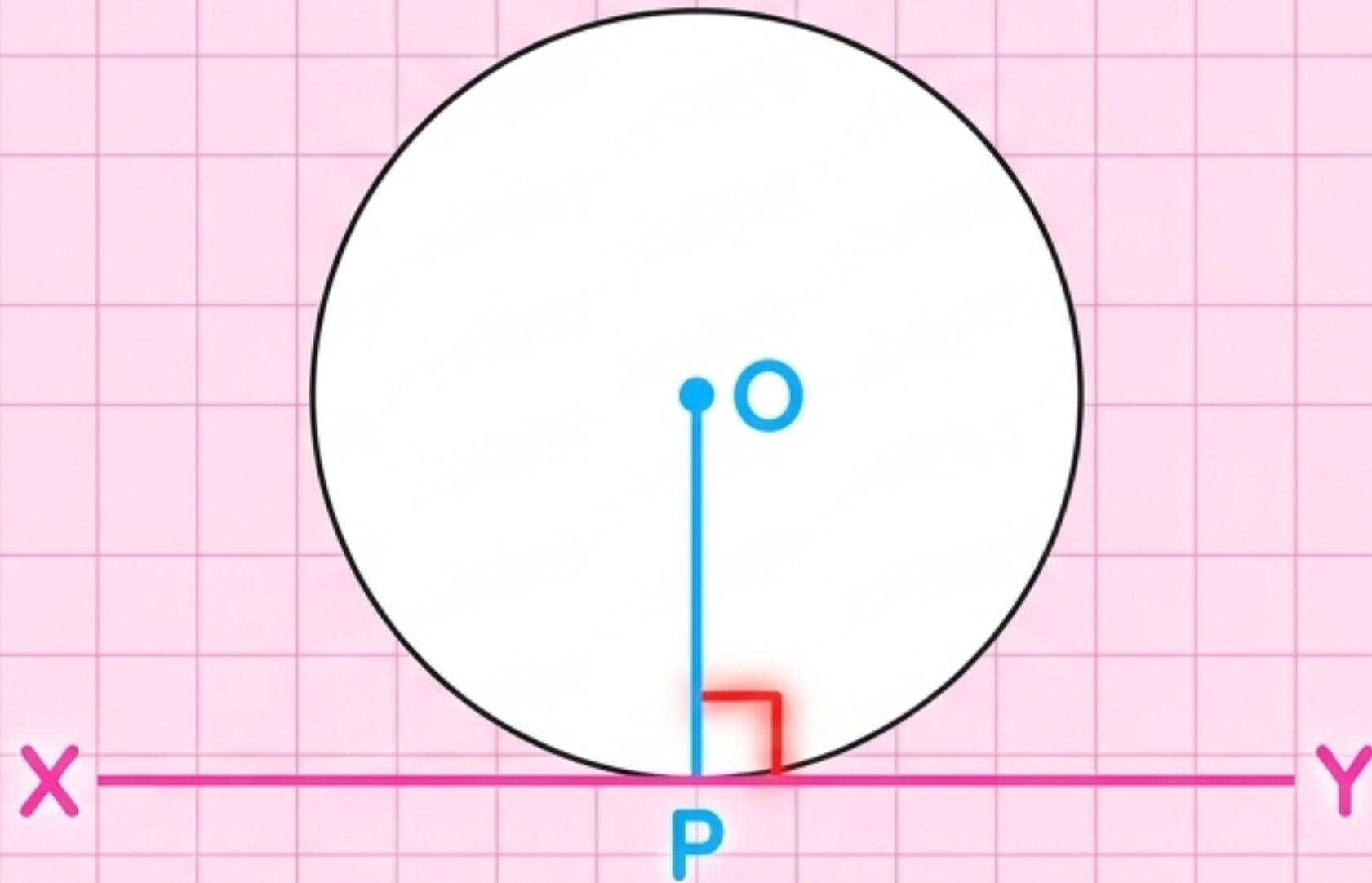
# The Point of Contact

The single common point where the tangent touches the circle.



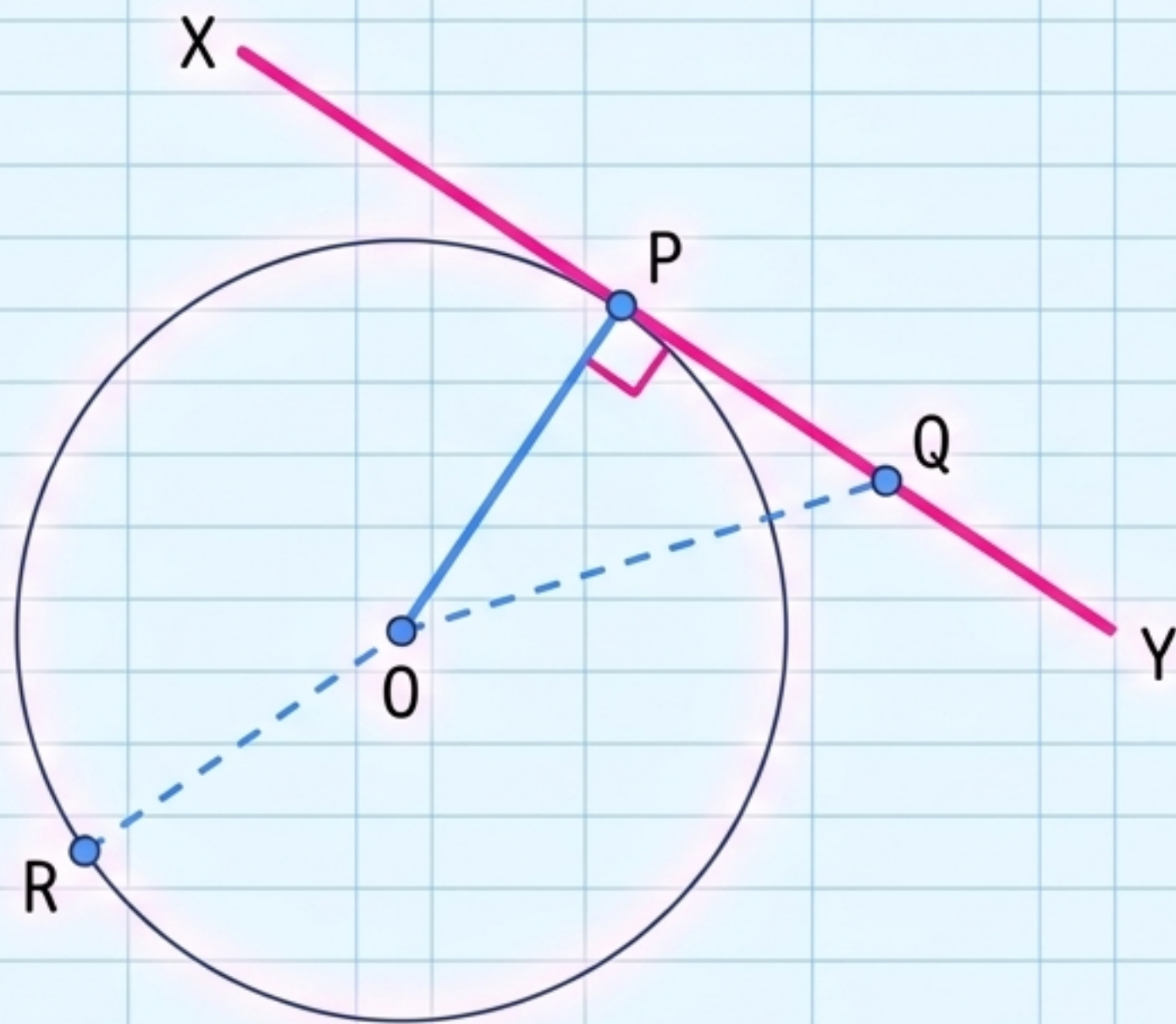
*The wheel moves along a line which is a tangent to the circle!*

# Theorem 10.1: The $90^\circ$ Rule



The tangent at any point of a circle is perpendicular to the radius through the point of contact.

# Visual Proof: The Shortest Path



Take any point Q  
on tangent XY  
(other than P).  
Q MUST be outside  
the circle.

Therefore, distance  
OQ is longer than  
radius OP.  
( $OQ > OP$ )

Since OP is the  
shortest of all  
distances from O  
to XY, it MUST be  
perpendicular!



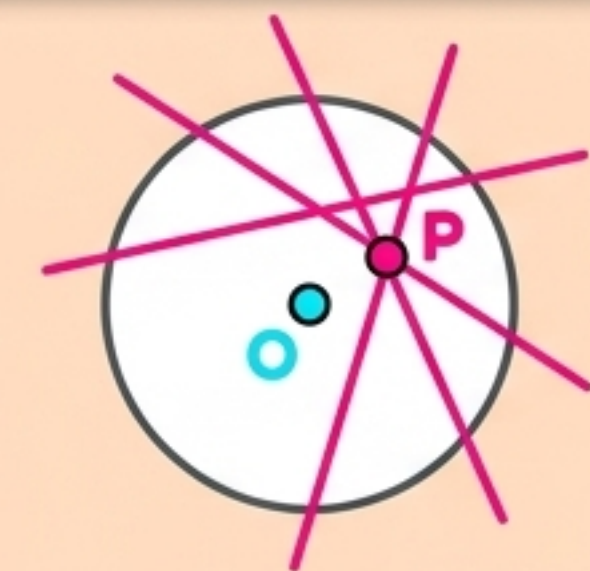
# How many tangents can we draw?

Where is the point located?

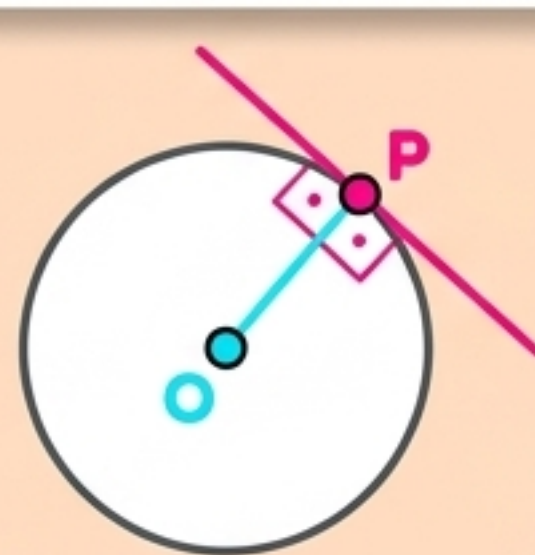
Inside

Outside

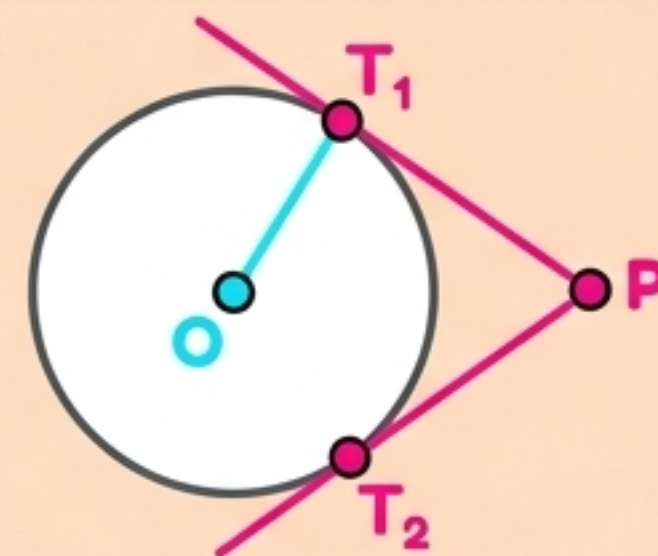
On the Edge



**0 Tangents**



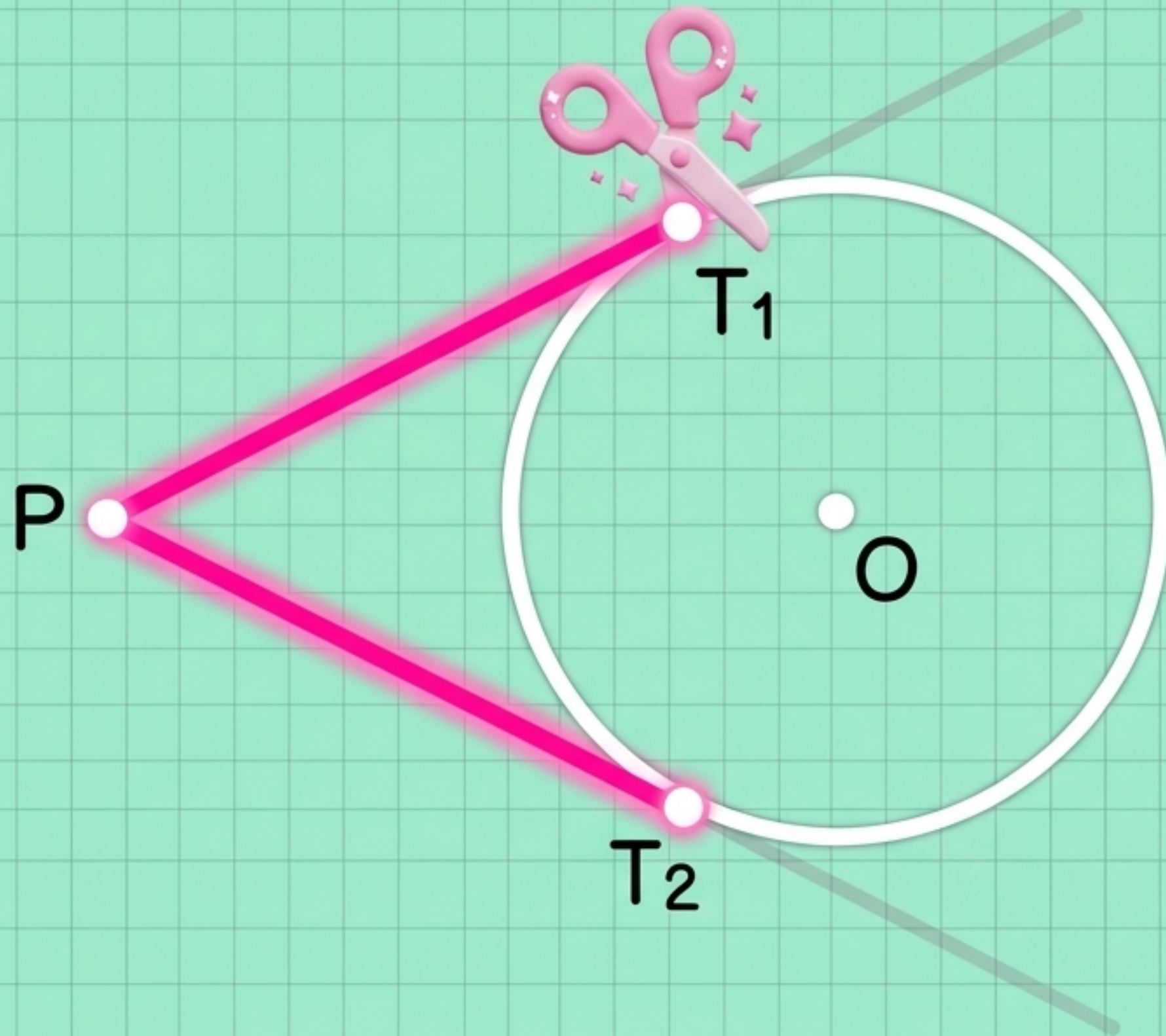
**Exactly 1 Tangent**



**Exactly 2 Tangents**



# Visualizing Tangent Lengths

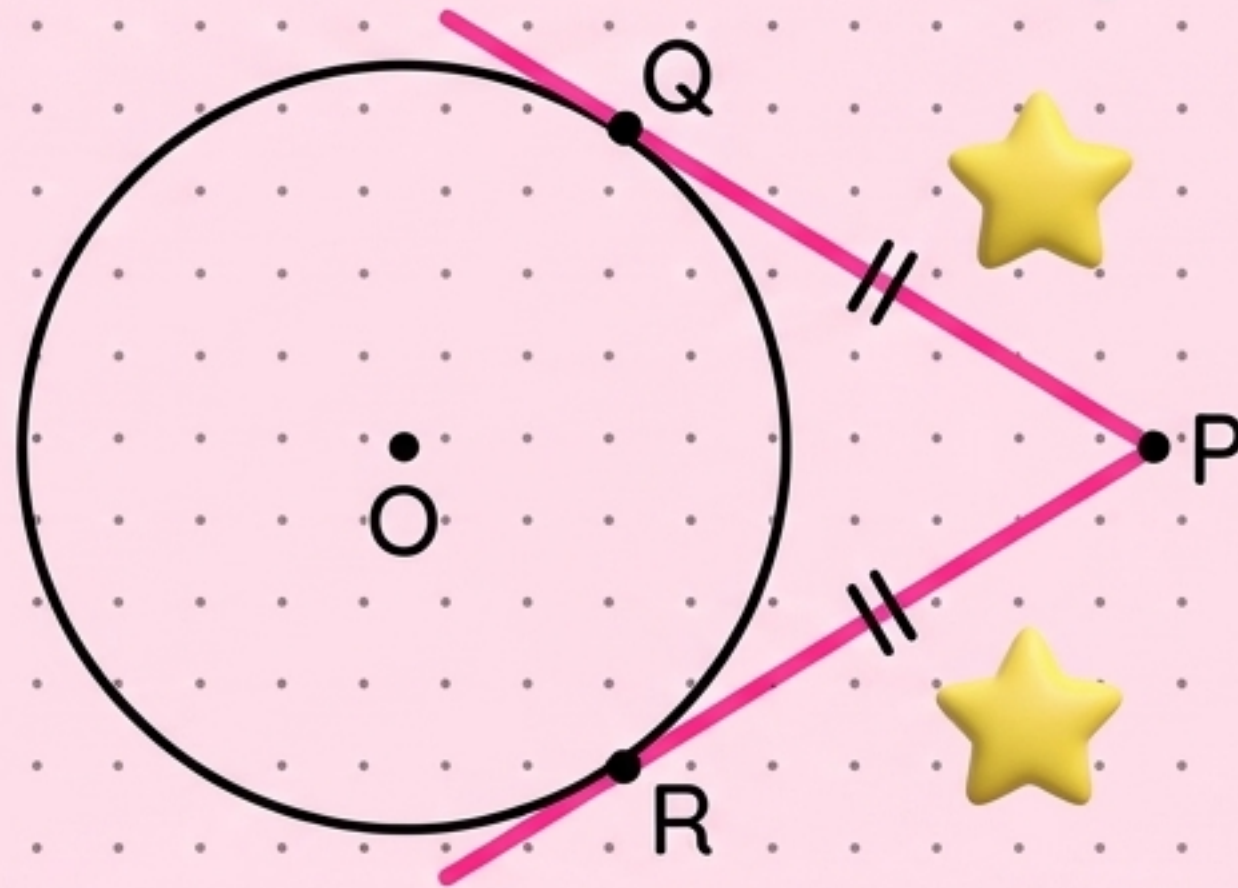


## Length of the Tangent

The length of the segment from the external point  $P$  to the exact point of contact with the circle.

Warning: Don't measure past the point of contact! The line goes on forever, but the \*length\* stops at the circle.

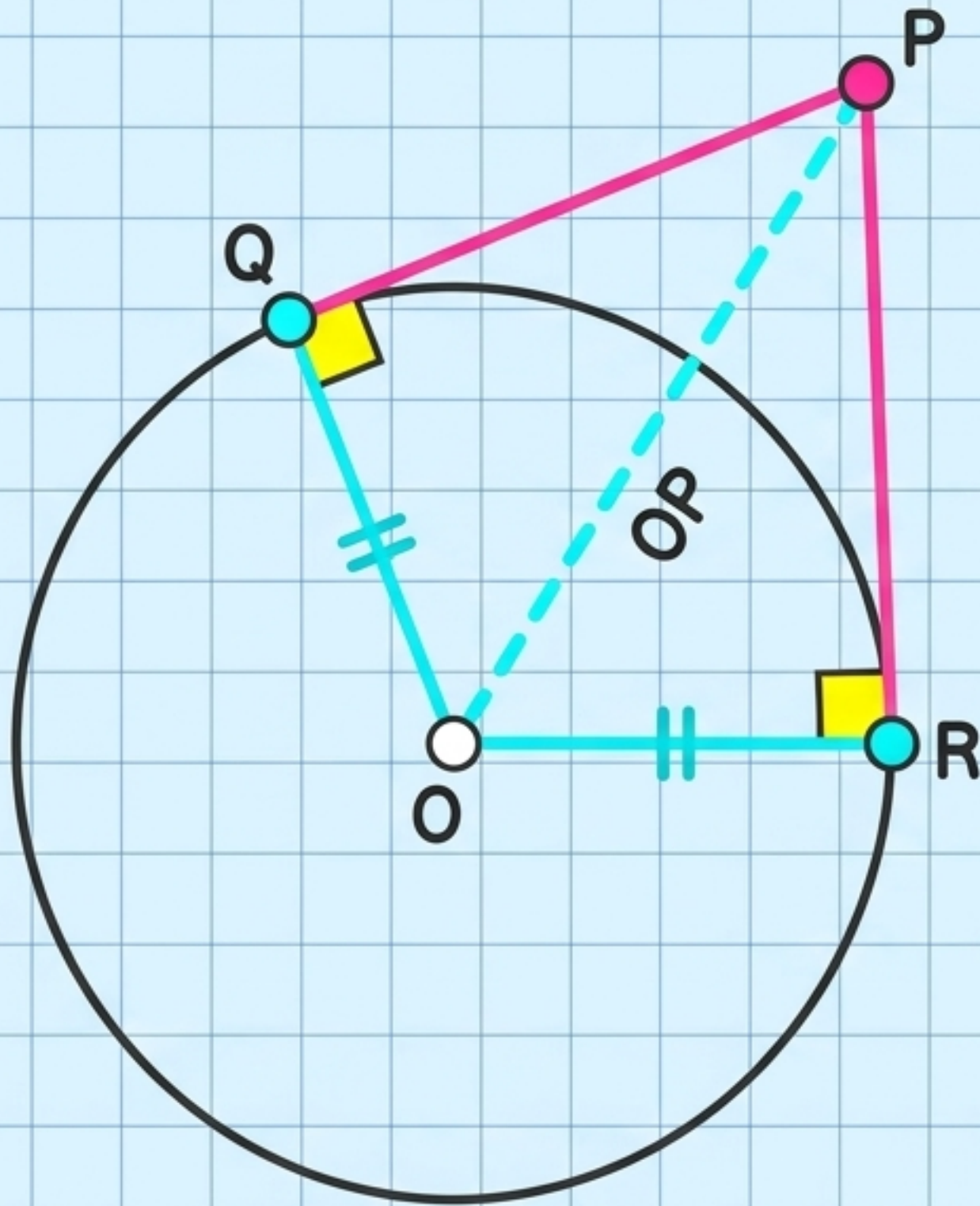
# Theorem 10.2: The Equal Twins



The lengths of tangents drawn from an external point to a circle are **equal**.

$$(PQ = PR)$$

# Visual Proof: The Mirror Image



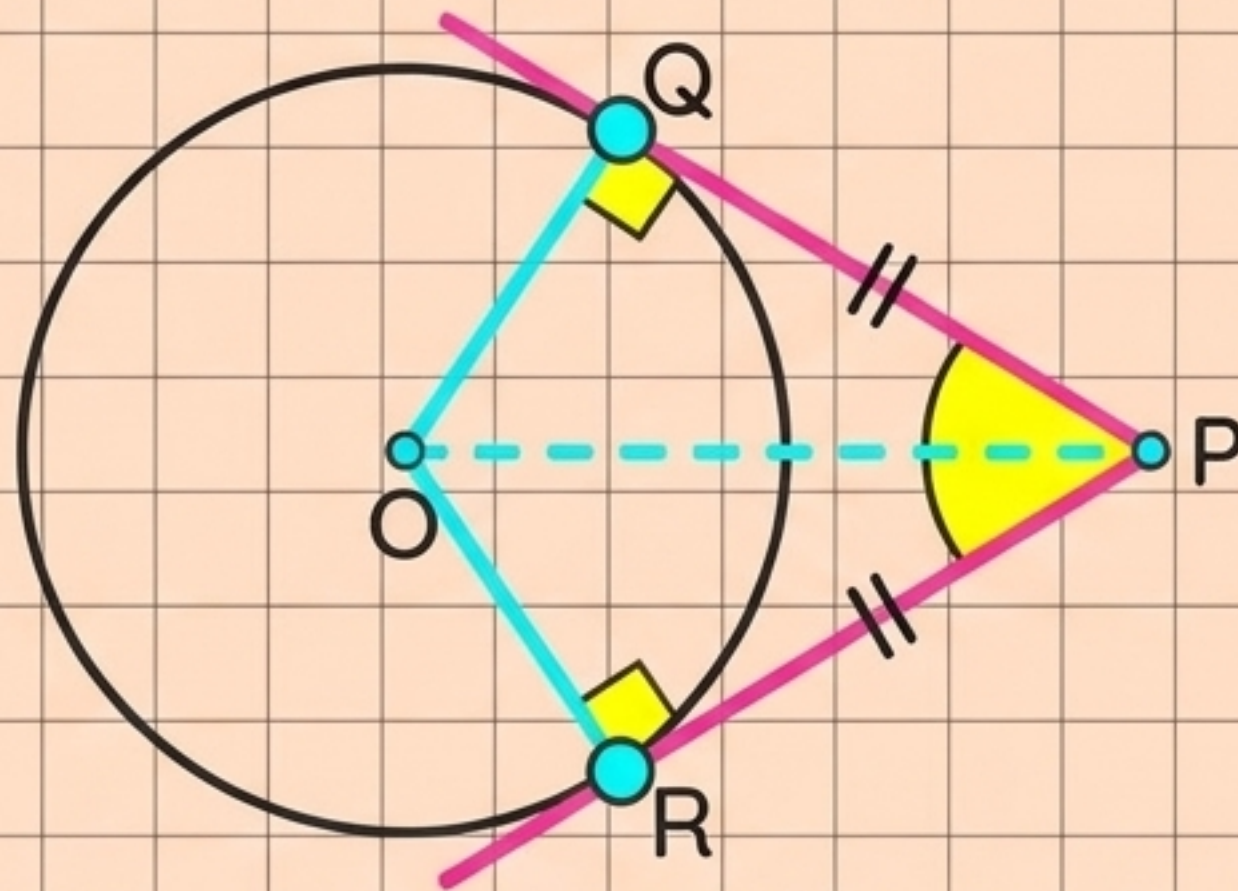
- ✓  $OQ = OR$  (Radii of same circle)
- ✓  $OP = OP$  (Shared common side)
- ✓  $\angle OQP = \angle ORP = 90^\circ$   
(Thanks to Theorem 10.1!)

**$\triangle OQP \cong \triangle ORP$  by  
RHS Congruence!**

**Therefore,  
 $PQ = PR$  (CPCT).**



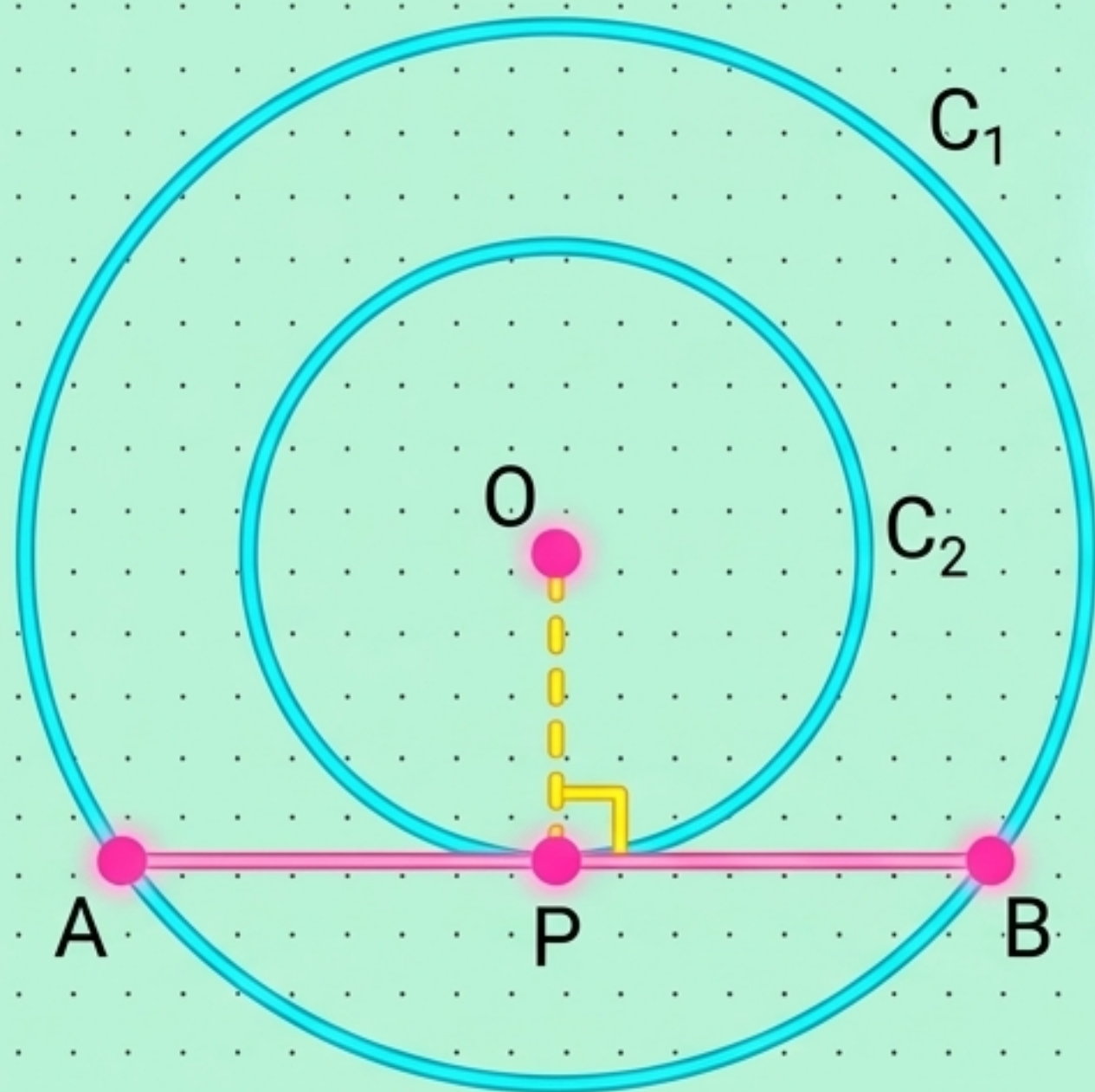
# Pro-Tip: The Angle Bisector Bonus!



Because the mirror triangles are perfectly equal, the center line (OP) perfectly slices the angle between the two tangents in half!

$$\angle OPQ = \angle OPR$$

# Application: Concentric Circles



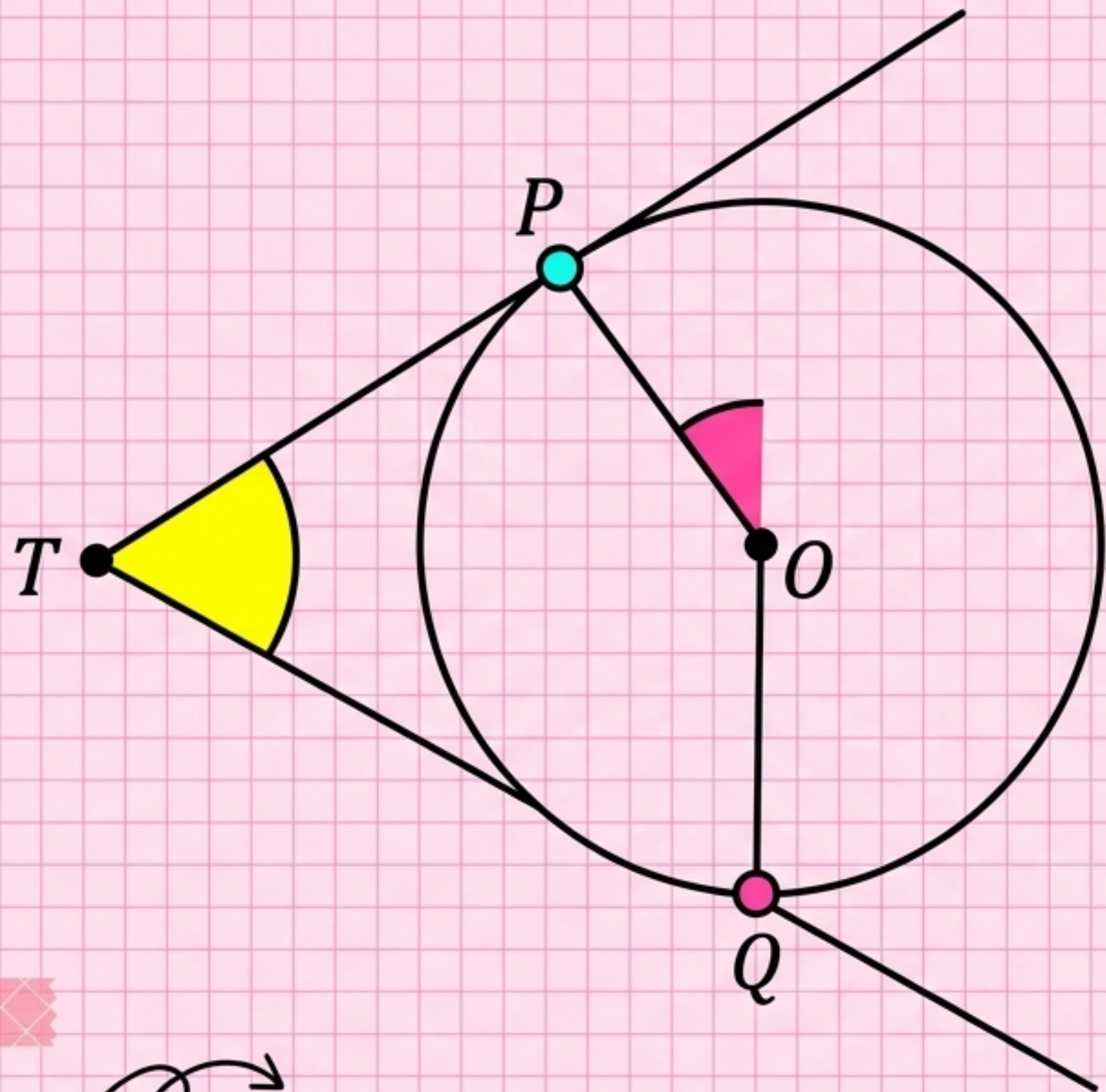
**Step 1:** AB is a tangent to inner circle  $C_2$ .  
So,  $OP \perp AB$  (Theorem 10.1).

**Step 2:** AB is a chord of outer circle  $C_1$ .

**Step 3:** A perpendicular from the center bisects a chord! (Class 9 throwback).

*Result:  $AP = BP$ .  
The chord is  
perfectly cut in half!*

# Puzzle Solved: Angles & Tangents



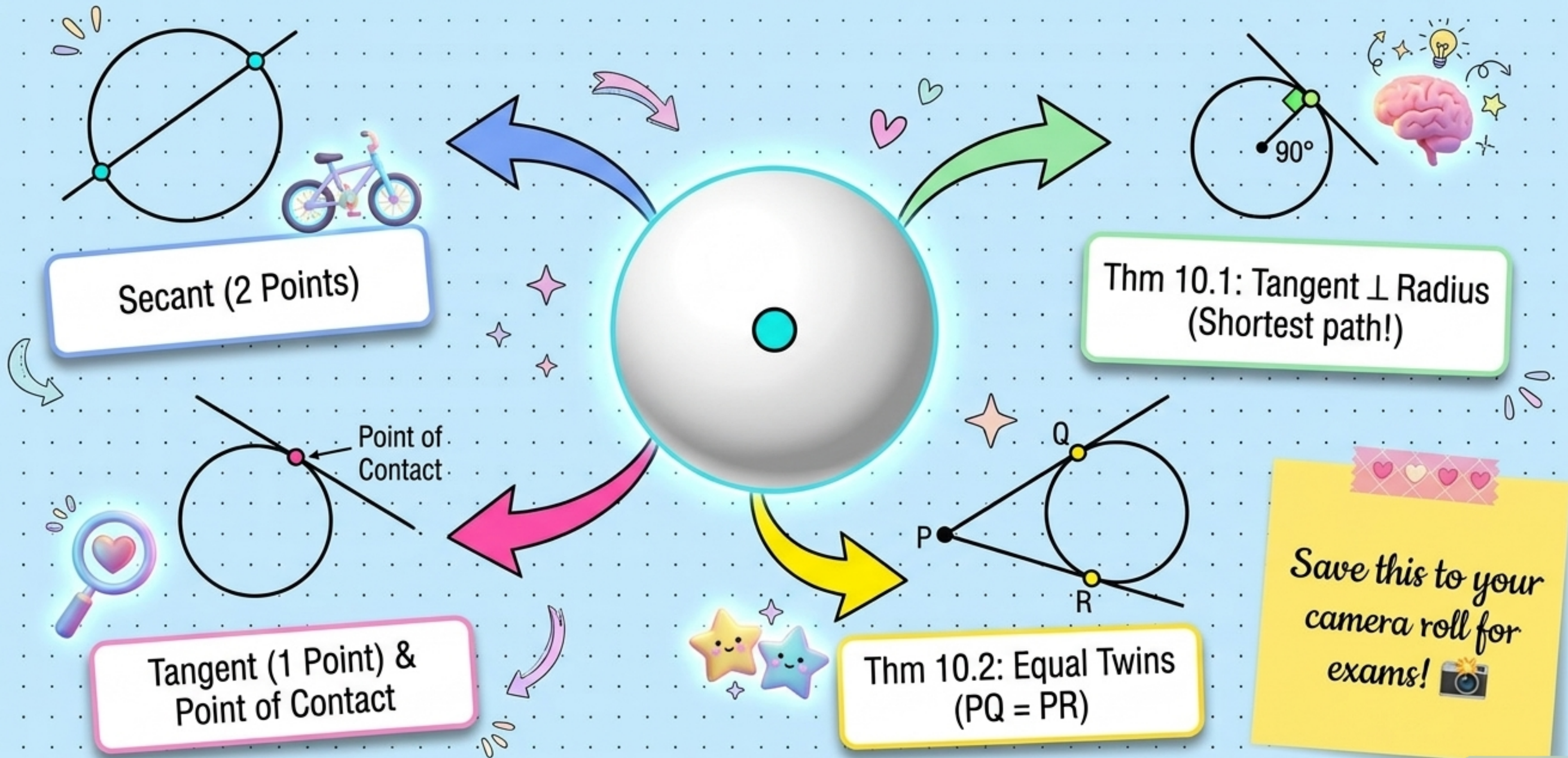
$TP = TQ$  (Thm 10.2).  
So,  $\triangle TPQ$  is isosceles!

This means base angles are equal:  
 $\angle TPQ = \frac{1}{2}(180^\circ - \angle PTQ)$ .

Radius is  $\perp$  to tangent (Thm 10.1).  
So  $\angle OPT = 90^\circ$ .

$$\angle PTQ = 2 \times \angle POQ!$$

# The Master Circle Cheat Sheet



Save this to your  
camera roll for  
exams! 📷